

Arm® Cortex®-M23
32-bit Microcontroller

M251/M252/M254/M256/M258 Series
CMSIS BSP
Revision History

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Revision 3.02.010 (Released 2026-09-16)

- Add SBOM.
- CMSIS upgrades to 6.3.0 and adds CMSIS-Compiler 2.2.0.
- Standard driver
 - PWM: PWM_EnableFaultBrake removes support of multiple brake sources.
- Sample code
 - ISP_DFU: Remove dfu-util.
- ThirdParty
 - FreeRTOS: Upgrade to v11.3.0.
- Tool
 - HIDTransferTest
 - Move from USB_D sample codes.
 - Add Control transfer.

Revision 3.02.009 (Released 2026-04-29)

- Standard driver
 - FMC: FMC_WriteMultiple supports resume for timeout recovery.
 - PWM: PWM_EnableFaultBrake supports multiple brake sources.
- Sample code
 - IAR projects upgrade to use IAR 8.42.
 - ISP samples: Fix flash select setting in flash download of Keil project.
 - QSPI_DualMode_Flash, QSPI_QuadMode_Flash, SPI_Flash: Improve performance.
- Tool
 - TouchView: Fix TK index issue.

Revision 3.02.008 (Released 2025-10-08)

- Standard driver
 - I2C: Add STOP bit check for I2C transmission completely.
 - PDMA
 - PDMA_INT_TRANS_DONE, PDMA_INT_EMPTY, PDMA_INT_TIMEOUT, PDMA_DisableInt, PDMA_EnableInt: Support multi-bit mask.
 - PDMA_INT_ALIGN: Add.
 - SPI: SPI_DISABLE_3WIRE_MODE and SPI_ENABLE_3WIRE_MODE add.
- Sample code
 - BPWM_DoubleBuffer, PWM_DeadTime, PWM_DoubleBuffer, TIMER_PWM_ChangeDuty: IRQ handler clears interrupt flag at first.
 - EADC_AverageCMP, QSPI_PDMA_Master, QSPI_PDMA_Slave: Add.

- ISP_I2C and ISP_SPI: Fix potential command failure.
- USB_D_HID_RemoteWakeup: Fix host cannot sleep issue.
- VS Code projects upgrade to use GCC 14.3.1.
- Tool
 - TouchView and TouchView_MP replace NuSenadj and NuSenadj_MP.

Revision 3.02.007 (Released 2024-09-25)

- CMSIS: Upgrade to v6.1.0.
- Keil startup_M251.c: Instead of ASM code.
- Standard driver
 - UART: UART_RESET_RXFIFO and UART_RESET_TXFIFO add.
- Sample code
 - Add VS Code projects with AC6/GCC toolchain.
 - I2C_Loopback_10bit: Merge into I2C_Loopback.
 - I2C_Master_10bit: Merge into I2C_Master.
 - I2C_Slave_10bit: Merge into I2C_Slave.
 - ISP_UART: Reset RX FIFO after response sent.
 - USB_D_Audio_Headset: Add.
 - USB_D_Mass_Storage_SRAM: Reserve memory space for mass storage instead of direct addressing.

Revision 3.02.006 (Released 2024-03-21)

- Peripheral header
 - rtc_reg.h: Fix RTC_CLKFMT_DCOMPEN_Pos.
- Standard driver
 - retarget.c: Fix __ARM_use_no_argv multiple definition issue in newer Keil versions.
 - BPWM: BPWM_ConfigOutputChannel fixes duty issue.
 - CLK
 - CLK_PowerDown clears CPU deep sleep mode select after wakeup.
 - CLK_SysTickDelay improves robustness.
 - I2C: I2C_WAIT_READY fixes potential blocking issue.
 - LCD: LCD_SetPixel and LCD_SetAllPixels fix segment count issue.
 - PWM: PWM_ConfigOutputChannel fixes duty issue.
 - TIMER_PWM: TPWM_ConfigOutputFreqAndDuty fixes duty issue.
- Sample code

- BPWM_SwitchDuty, PWM_SwitchDuty: Fix 100% duty issue.
- I2C_Master, I2C_Master_10bit: Add I2C_DisableTimeout().
- ISP_DFU, ISP_MSC: Reduce code size.
- NK_M258KE_LCD_TK: Support NuMaker M258KE board V1.1 and above.
- QSPI_DualMode_Flash, QSPI_QuadMode_Flash, SPI_Flash: Support more Winbond SPI flashes.
- SYS_PowerDown_MinCurrent: Remove LVR disable setting.
- USB_D samples: Remove LPM support.
- USB_D HID samples: Support LED status.

Revision 3.02.005 (Released 2023-08-07)

- Library
 - LDROM.icf and LDROM.sct: Fix LDROM address.
- Sample code
 - ISP samples: Fix LDROM download failure in IAR and Keil.
 - ISP_DFU: Update Windows driver with Nuvoton vendor name.
 - NK_M258KG_LCD_TK: Improve touch key sensitivity.
 - USB_D samples: Remove EP1 stall while setup error occurs.

Revision 3.02.004 (Released 2023-02-06)

- Library
 - TKLib: Add hysteresis recover mechanism for baseline tracking.
- Sample code
 - NK_M256SD_LCD_TK and NK_M258KE_LCD_TK: TK_SE_NUM add.
 - SYS_PowerDown_MinCurrent: Add.
 - USB_D samples: Avoid wakeup failure.

Revision 3.02.003 (Released 2022-07-21)

- Standard driver
 - retarget.c: Fix print issue with IAR EWARM v9.20.
 - CLK
 - CLK_PowerDown adds check of LIRC disable for DPD mode.
 - CLK_WaitClockDisable adds.
 - I2C: Error codes add.
 - LCD
 - LCD output control constant definitions add.

- LCD_SetSavingMode adds.
- SYS: SYS_GPA_MFPH_PA8MFP_DAC0_OUT and SYS_GPA_MFPH_PA9MFP_DAC1_OUT.
- UART: UART_RX_IDEL is replaced by UART_RX_IDLE.
- Library
 - SmartcardLib: Support cards not compatible with ISO-7816.
- Sample code
 - ACMP_CompareDAC: Add external negative source from DAC voltage output.
 - DAC samples: Extend DAC output pin to support more chips.
 - FreeRTOS Blinky sample: Add.
 - I2C samples: Improve error handling.
 - I2C_Loopback_10bit, I2C_Master_10bit, I2C_Slave_10bit, LCD_PowerConsumption, TK_PowerDownMode, and Level1_Training samples: Add.
 - ISP_DFU, ISP_HID, and ISP_MSC: Enable USB trim function.
 - ISP_SPI: Add GCC project.
 - ISP samples
 - Add IAR projects.
 - Add scatter file to check image size.
 - Enable function section linking and image size optimization in Keil projects.
 - NK_M256SD_LCD_TK and NuMaker-M256SD LCD samples: Add.
 - NK_M258KE_LCD_TK, NK_M258KG_LCD_TK, and LCD samples: Add LCD output setting.
 - RTC samples: Skip time setting if RTC initialized.
 - RTC_Alarm_Wakeup: Move to PowerManagement folder.
 - SYS_DPDMODE_Wakeup: Add wait for LIRC disable.
 - SYS_DPDMODE_Wakeup and SYS_PowerDownMode: Add reset status check.
 - UART_PDMA: Fix compiling issue of polling method.
- ThirdParty
 - FreeRTOS: Add.

Revision 3.02.002 (Released 2022-03-15)

- Library
 - LCDLib: Synchronize with NuTool-LCDView tool.
 - TKLib: Improve anti-noise capability.
- Sample code

- DAC_GroupMode, EADC_BPWM_Trigger, EADC_PDMA_BPWM_Trigger, and NK_M258KG_LCD_TK: Add.
- LCD_Blinking and LCD_Pixel_OnOff: Use default Charge Pump setting.
- LCD samples: Use new LCDLib.
- Tool
 - NuSenadj_MP: Update.

Revision 3.02.001 (Released 2022-01-20)

- Support CRYPTO.
- Peripheral header
 - dac_reg.h: Fix DAC_CTL_GRPEN_Msk.
 - lcd_reg.h: Update base on TRM.
 - tk_reg.h: Support up to 24 keys.
- Standard driver
 - Improve infinite loop prevention with timeout counter and error code.
 - BPWM: BPWM_ConfigCaptureChannel and BPWM_ConfigOutputChannel fix wrong clock for chips without PLL.
 - CLK: CLK_EnableSysTick correctly uses external reference clock.
 - DAC: DAC_SetDelayTime fixes calculation overflow issue.
 - GPIO: PF15 adds.
 - LCD
 - LCD_GET_CHARGE_TIME and LCD_WAVEFORM_TYPE fix.
 - LCD_SET_CONTINUOUS_FRAME and LCD_SET_NULL_FRAME_TIME remove.
 - LCD_SET_PAUSE_TIME and LCD_SET_PAUSE_TYPE add.
 - PWM: PWM_ConfigCaptureChannel and PWM_ConfigOutputChannel fix wrong clock for chips without PLL.
 - RTC: RTC_Open adds return value.
 - SCUART: SCUART_Write adds return value.
 - SPI: Support SPI1.
 - TIMER: TIMER_Delay and TIMER_ResetCounter add return value.
 - TK: Support up to 24 keys.
- Library
 - TKLib: Support up to 24 keys.
- Sample code
 - EADC_Timer_Trigger: Fix wrong timer period.
 - I2C samples: Remove redundant code.

- ISP_I2C, ISP_RS485, and ISP_UART: Add GCC project.
- ISP_MSC: Add.
- SPI_HalfDuplex and SPI_PDMA_LoopTest: QSPI0 is replaced by USPI0.
- USB_D_HID samples: Unify INT_IN interval.
- USB_D_Mass_Storage_CDROM: Fix Linux and Mac compatibility issue.
- Tool
 - NuSenadj: Update.

Revision 3.01.002 (Released 2021-06-25)

- system_M251.h: __HXT and __LXT add redefinition check.
- Peripheral header
 - psio_reg.h: A reserved member adds to GNCT_T structure.
- Standard driver
 - SYS: PF.15 MFP adds.
- Library
 - LCDLib
 - Fix display flicker.
 - Variable legality check for LCD print and putchar.
 - TKLib
 - Add all key calibration for power-down mode, store address scan, and store pin selection to Flash.
 - Decrease parameters Flash size.
 - Fix reference CB overflow, power-on fine tune.
- Sample code
 - BPWM_Capture: PWM0 channel 0 is replaced by Timer0.
 - I2C and USCI_I2C samples: Enable schmitt trigger of I²C pins.
 - ISP_HID, ISP_I2C, ISP_RS485, and ISP_UART: Enable GPIO and SysTick clocks.
 - NuMaker-M258KE: Support CKO shielding, default calibration parameters, and null frame in power-down mode.
 - SPI_SlaveFIFOmodeINT, USCI_SPI_SlaveModeINT: Add.
 - TIMER_FreeCountingMode: Save the first captured data.
 - UART_Wakeup: Provide more wake-up methods.
 - USB_D_Audio_Codec: Refactor.
 - USB_D_MassStorage_CDROM: Remove 2 KB file size limitation.
 - USB_D_VCOM samples: Fix potential UART Tx FIFO overflow issue.
 - USB_D_VCOM_DauiPort: Fix power-down blocked issue and wrong baud rate issue.

■ Tool

- NuSenadj: Move to TK subfolder.
- NuSenadj_MP: Add.

Revision 3.01.001 (Released 2020-11-16)

■ Support M254/M256/M258 series.

■ Peripheral header

- lcd_reg.h, tk_reg.h: Add.
- eadc_reg.h: EADC_T.OFFSETCAL, EADC_CTL_CALEN_Msk, EADC_CTL_CALEN_Pos, EADC_OFFSETCAL_OFFSETCANCEL_Msk, and EADC_OFFSETCAL_OFFSETCANCEL_Pos remove.

■ Standard driver

- retarget.c: Fix program terminated issue with optimization level 0 of Arm Compiler 6.
- EADC
 - EADC_ENABLE_CMP0, EADC_ENABLE_CMP1, EADC_ENABLE_CMP2, and EADC_ENABLE_CMP3 fix bits overwritten issue.
 - EADC_DISABLE_OFFSETCANCEL and EADC_ENABLE_OFFSETCANCEL remove.
- LCD, TK: Add.
- QSPI: QSPI_DISABLE_TX_RX_PDMA and QSPI_TRIGGER_TX_RX_PDMA add.
- SPI: SPI_DISABLE_TX_RX_PDMA and SPI_TRIGGER_TX_RX_PDMA add.
- SYS: SYS_SetPowerLevel improves power level switch stability.
- USCI-SPI: USPI_DISABLE_TX_RX_PDMA and USPI_TRIGGER_TX_RX_PDMA add.
- USCI-UART: UART_PDMA_DISABLE and UART_PDMA_ENABLE add.

■ Library

- LCDLib, TKLib: Add.

■ Sample code

- EADC_BandGapCalculateAVDD, LCD_Blinking, LCD_Pixel_OnOff, LCD_Print_Text, NK_M258KE_LCD_TK, and USBD_BC12_HID_Mouse: Add.
- EADC_OffsetCancel: Remove.
- GPIO samples: Enable GPIO clock.
- PDMA_BasicMode: Fix miss word alignment.
- USBD_Audio_Codec: Fix noise and I²C timeout.
- USBD_VCOM_DualPort: Fix only one VCOM port in Linux.

Revision 3.00.005 (Released 2020-06-10)

- Add Apache-2.0 license declaration.
- Standard driver
 - retarget.c: SendChar fixes uninitialized pointer.
 - TIMER: TIMER_SET_OPMODE adds.
 - UART: UART_DEGLITCH_DISABLE, UART_DEGLITCH_ENABLE, UART_PDMA_DISABLE, and UART_PDMA_ENABLE add.
 - USCI-UART: UART_DEGLITCH_DISABLE and UART_DEGLITCH_ENABLE add.
- Sample code
 - CLK_ClockDetector: HCLK clock source uses HXT instead of PLL.
 - GPIO_SingleCycleIO, I2C_SMBus, MPU, SYS_PowerDownMode, Sys_PowerMode, and USB_D_CCID: Add.
 - QSPI_DualMode_Flash, QSPI_QuadMode_Flash, and SPI_Flash: Fix the 256th byte written issue.

Revision 3.00.004 (Released 2019-12-31)

- Remove CRYPTO.
- Standard driver
 - retarget.c: _write and SendChar_ToUART fix '\r' issue.
 - CRC: CRC_SET_WDATA_LEN adds.
 - FMC
 - FMC_Write128 adds note to run in SRAM instead of inline function.
 - FMC_WriteConfig fixes return value.
 - SYS: SYS_PLCTL_PLSEL_PL2 is replaced by SYS_PLCTL_PLSEL_PL3.
 - TIMER_PWM: TPWM_ConfigOutputFreqAndDuty fixes wrong configuration for 1 Hz.
- Sample code
 - FMC_MultiWordProgram: Use FMC driver API.
 - I2C_Wakeup_Slave: Improve wakeup robustness.
 - ISP_DFU: Add.
 - UART_LIN and UART_SingleWire: Fix '\r' issue.
 - USB_D samples: Improve USB compatibility.

Revision 3.00.003 (Released 2019-08-30)

- M251.h: Include core_cm23.h instead of core_armv8mbl.h.
- Peripheral header
 - clk_reg.h, rtc_reg.h: Update base on TRM.

■ Standard driver

- I2C: I2C_SetSlaveAddr and I2C_SetSlaveAddrMask support 10-bit slave address and error handling for new transfer.
- TIMER
 - TIMER_GetModuleClock uses correct PCLK.
 - TIMER_Open fixes return value.
- TIMER_PWM
 - TPWM_ClearWakeupFlag, TPWM_DisableTrigger, TPWM_DisableWakeup, TPWM_EnableTrigger, TPWM_EnableWakeup, TPWM_GetWakeupFlag, TPWM_CLEAR_TRG_DAC_STATUS, TPWM_CLEAR_TRG_PDMA_STATUS, TPWM_GET_TRG_DAC_STATUS, and TPWM_GET_TRG_PDMA_STATUS add.
 - TPWM_OUTPUT_TOGGLE, TPWM_OUTPUT_NOTHING, TPWM_OUTPUT_LOW, TPWM_OUTPUT_HIGH, TPWM_CLKSRC_HXT, TPWM_CLKSRC_LXT, TPWM_CLKSRC_PCLK, TPWM_CLKSRC_TX, TPWM_CLKSRC_LIRC, and TPWM_CLKSRC_HIRC remove.
- USCI_SPI: USPI_SET_SS_HIGH fixes implementation.

■ Library

- SmartcardLib: SCLIB_CheckCDEvent_ByVar adds.

■ Sample code

- EADC_BandGap: Show voltage result and modified extern sample time for band-gap.
- FMC_ExecInSRAM: Add GCC project.
- I2C_Master_PDMA, I2C_PDMA, and I2C_Slave_PDMA: Be replaced by I2C_PDMA_TRX.
- ISP samples, QSPI_Slave3Wire, and SPI_HalfDuplex: Add.
- SC_ReadATR and SC_ReadSimPhoneBook: Add software card detection.
- UART_TxRx_Function: Fix data lost.
- USB samples: Fix potential issue.
- USCI_SPI_Loopback, USCI_SPI_MasterMode, and USCI_SPI_SlaveMode: Add using GPIO pin instead of SS pin.

Revision 3.00.002 (Released 2019-02-22)

- system_M251.c: Set flash access cycle to 0x03.
- Standard driver
 - CLK: CLK_SET_PCLK_DIVIDER remove.
- Sample code
 - Use other clock source instead of HXT.

- SYS_TrimHIRC: Add auto trim disable.
- USBD_Audio_Codec: Clear buffer before play / record.

Revision 3.00.001 (Released 2018-12-25)

- Initial release.

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